# Partial Differential Equations

PENG

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## Contents



## 1 Four Fundamental Linear PDEs

**Definition 1.1.** *Four fundamental linear PDEs [1]:*

- *Transport equation*
- *Laplace's equation*
- *Heat equation*
- *Wave equation*

**Definition 1.2.** *Gradient vector:*

$$Du = \nabla u := \begin{bmatrix} \frac{\partial u}{\partial x_1} \\ \frac{\partial u}{\partial x_2} \\ \vdots \\ \frac{\partial u}{\partial x_n} \end{bmatrix} \tag{1.1}$$

**Definition 1.3.** *Vector dot product and matrix multiplication:*

$$A_{n\times 1} \cdot B_{n\times 1} = A^T B = [a_1 a_2 \dots a_n] \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} = \sum_{i=1}^{n} a_i b_i \tag{1.2}$$

**Lemma 1.4.**

$$\frac{dy}{d(x+b)} = \frac{dy}{dx} \tag{1.3}$$

*Proof.*

$$\frac{dy}{d(x+b)} = \frac{\frac{dy}{dx}}{\frac{d(x+b)}{dx}} = \frac{dy}{dx}$$

□

## 1.1 Transport Equation

**Definition 1.5.** *Transport equation:*

$$u_t + b \cdot Du = \frac{\partial u}{\partial t} + [b_1 b_2 \dots b_n] \begin{bmatrix} \frac{\partial u}{\partial x_1} \\ \frac{\partial u}{\partial x_2} \\ \vdots \\ \frac{\partial u}{\partial x_n} \end{bmatrix} = \frac{\partial u}{\partial t} + \sum_{i=n}^{n} b_i \frac{\partial u}{\partial x_i} = 0 \tag{1.4}$$

**Example 1.6.** *For 1D and 2D cases:*

$$\frac{\partial u}{\partial t} + b \frac{\partial u}{\partial x} = 0$$
$$\frac{\partial u}{\partial t} + b_1 \frac{\partial u}{\partial x} + b_2 \frac{\partial u}{\partial y} = 0$$

**Theorem 1.7.** *For Eq. 1.4, $u$ is constant on the line $(x,t) + (b,1)s$ for $s \in \mathbb{R}$.*

*Proof.* Let $z(s) := u(x + bs, t + s)$.

$$\frac{dz}{ds} = \frac{\partial u}{\partial (t+s)} \frac{d(t+s)}{ds} + \frac{\partial u}{\partial (x+sb)} \frac{d(x+sb)}{ds} = \frac{\partial u}{\partial t} + b \frac{\partial u}{\partial x} \overset{\text{Eq. 1.4}}{=} 0$$
$$z(s) = u(x + sb, t + s) = C$$

□

**Proposition 1.8.** *Initial-value problem:*

$$\begin{cases} \frac{\partial u}{\partial t} + b \frac{\partial u}{\partial x} = 0 \\ u(x,0) = g(x) \end{cases}$$

$$u(x,t) \overset{Theo.\ 1.7}{=} u(x - tb, 0) = g(x - tb)$$

**Proposition 1.9.** *Nonhomogenous problem:*

$$\begin{cases} \frac{\partial u}{\partial t} + b \cdot \frac{\partial u}{\partial x} = f(x,t) \\ u(x,0) = g(x) \end{cases}$$

*Let $z(s) := u(x + sb, t + s)$.*

$$\frac{dz}{ds} = f(x + sb, t + s)$$
$$z(s) = u(x + sb, t + s) = \int f(x + sb, t + s) ds$$
$$z(0) - z(-t) = u(x,t) - u(x - tb, 0) = u(x,t) - g(x - tb) = \int_{-t}^{0} f(x + sb, t + s) ds$$
$$u(x,t) = g(x - tb) + \int_{-t}^{0} f(x + sb, t + s) ds$$

## 1.2 Laplace's Equation

**Definition 1.10.** *Laplace's equation:*

$$\Delta u = \nabla \cdot \nabla u = \nabla^2 u = \left[ \frac{\partial}{\partial x_1} \frac{\partial}{\partial x_2} \cdots \frac{\partial}{\partial x_n} \right] \begin{bmatrix} \frac{\partial u}{\partial x_1} \\ \frac{\partial u}{\partial x_2} \\ \vdots \\ \frac{\partial u}{\partial x_n} \end{bmatrix} = \sum_{i=1}^{n} \frac{\partial^2 u}{\partial x_i^2} = 0 \tag{1.5}$$

**Example 1.11.** *For 1D and 2D cases:*

$$\frac{\partial^2 u}{\partial x^2} = 0$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

**Definition 1.12.** *Outward derivative:*

$$\frac{\partial u}{\partial v} := \boldsymbol{v} \cdot Du = [v_1 v_2 \dots v_n] \begin{bmatrix} \frac{\partial u}{\partial x_1} \\ \frac{\partial u}{\partial x_2} \\ \vdots \\ \frac{\partial u}{\partial x_n} \end{bmatrix} = \sum_{i=1}^{n} v_i \frac{\partial u}{\partial x_i}$$

*where $\boldsymbol{v}$ is the unit outer normal vector of $\partial U$.*

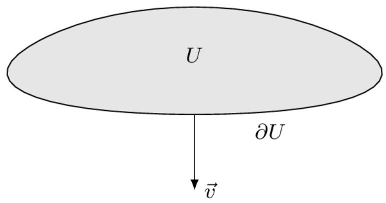


Figure 1.1: Outward derivative.

**Example 1.13.** *For 1D and 2D cases:*

$$\frac{\partial u}{\partial v} = v \frac{\partial u}{\partial x}$$

$$\frac{\partial u}{\partial v} = v_1 \frac{\partial u}{\partial x} + v_2 \frac{\partial u}{\partial y}$$

**Theorem 1.14.** *Gauss-Green theorem:*

$$\begin{aligned} \int_U \nabla \cdot \boldsymbol{u} dx = \int_U \left[ \frac{\partial}{\partial x_1} \frac{\partial}{\partial x_2} \cdots \frac{\partial}{\partial x_n} \right] \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} dx = \int_U \sum_{i=1}^{n} \frac{\partial u_i}{\partial x_i} dx \\ = \int_{\partial U} \boldsymbol{u} \cdot \boldsymbol{v} dS = \int_{\partial U} [u_1 u_2 \dots u_3] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} dS = \int_{\partial U} \sum_{i=1}^{n} u_i v_i dS \end{aligned} \tag{1.6}$$

**Example 1.15.** *For 1D and 2D cases:*

$$\int_U \frac{\partial u}{\partial x} dx = \int_{\partial U} uvdS$$

$$\int_U \frac{\partial u_1}{\partial x} + \frac{\partial u_2}{\partial y} dx = \int_{\partial U} u_1 v_1 + u_2 v_2 dS$$

Fundamental solution:

$$u(x) = v(r)$$

$$r = |x| = \sqrt{\sum_{i=1}^{n} x_i^2}$$

$$\frac{\partial r}{\partial x_i} = \frac{x_i}{r}$$

$$\frac{\partial u}{\partial x_i} = \frac{dv}{dr}\frac{x_i}{r}$$

$$\frac{\partial^2 u}{\partial x_i^2} = \frac{d^2 v}{dr^2}\frac{x_i^2}{r^2} + \frac{dv}{dr}\frac{r^2 - x_i^2}{r^3}$$

$$\Delta u = \sum_{i=1}^{n} \frac{\partial^2 u}{\partial x_i^2} = \frac{d^2 v}{dr^2} + \frac{dv}{dr}\frac{n-1}{r} = 0$$

For $n = 1$:

$$\frac{d^2 v}{dr^2} = 0$$

$$v(r) = r + C$$

For $n \geq 2$:

$$\frac{d^2 v}{dr^2} + \frac{dv}{dr}\frac{n-1}{r} = 0$$

$$\frac{v''(r)}{v'(r)} = \ln\left(|v'(r)|\right)' = \frac{1-n}{r}$$

$$\ln\left(|v'(r)|\right) = (1-n)\ln(r) + \ln(C) = \ln\left(\frac{C}{r^{n-1}}\right)$$

$$|v'(r)| = \frac{C}{r^{n-1}}$$

$$v'(r) = \frac{C}{r^{n-1}}$$

For $n = 2$:

$$v'(r) = \frac{C}{r}$$

$$v(r) = C_1 \ln(r) + C_2$$

For $n \geq 3$:

$$v(r) = \frac{C_1}{2-n}\frac{1}{r^{n-2}} + C_2 = \frac{C_1}{r^{n-2}} + C_2$$

**Definition 1.16.** *Fundamental solution of Laplace's equation:*

$$\Phi(|x|) := \begin{cases} -\frac{1}{2\pi}\ln|x| & (n = 2) \\ \frac{1}{n(n-2)\alpha(n)}\frac{1}{|x|^{n-2}} & (n \geq 3) \end{cases} \tag{1.7}$$

**Definition 1.17.** *Poisson's equation:*

$$-\Delta u = -\sum_{i=1}^{n} \frac{\partial^2 u}{\partial x_i^2} = f \tag{1.8}$$

**Theorem 1.18.** *Solution of Poisson's equation:*

$$u(x) = \int_{y\in\mathbb{R}^n} \Phi(|x-y|) f(y)dy = \begin{cases} -\frac{1}{2\pi}\int_{\mathbb{R}^n} \ln\left(|x-y|\right) f(y)dy & (n = 2) \\ \frac{1}{n(n-2)\alpha(n)}\int_{\mathbb{R}^n} \frac{f(y)}{|x-y|^{n-2}}dy & (n \geq 3) \end{cases} \tag{1.9}$$

# 2 Asymptotics

## 2.1 Singular Perturbations

# References

[1] Lawrence C Evans. *Partial differential equations*. Vol. 19. American mathematical society, 2022. ISBN: 1470469421.